

**Why Risk it, When You Can {rix} it: A Tutorial for Computational Reproducibility Focused  
on Simulation Studies**

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### Abstract

Computational reproducibility remains limited in psychological research, despite widespread norms for sharing data and analysis code. One reason is that reproducibility exists on a continuum, ranging from partial transparency—such as providing scripts or software version numbers—to fully executable research compendia that regenerate all results from raw code. In this article, we introduce Nix and the {rix} R package as a practical framework for achieving full computational reproducibility in simulation-based research. We provide a step-by-step tutorial demonstrating how {rix} can be used to define, build, and share isolated, project-specific software environments that precisely capture R versions, package dependencies, system libraries, and integrated development environments. We further illustrate this workflow by reproducing a complete manuscript using Quarto and the {apaquarto} extension, showing how analyses, figures, and text can be regenerated in a single, executable pipeline. Together, these tools lower the technical barrier to robust, end-to-end reproducibility and offer a scalable solution for simulation studies and methodological research in psychology and related fields.

*Keywords:* reproducibility, Nix, simulation studies, R, computational methods

## Why Risk it, When You Can {rix} it: A Tutorial for Computational Reproducibility Focused on Simulation Studies

Psychological science is in the midst of a credibility revolution ([Vazire, 2018](#)), yet much of that scrutiny has focused on empirical studies. Monte Carlo simulation studies (hereafter referred to as simulation studies) have received far less ([Siepe et al., 2024](#)), even though they are a primary tool for evaluating the statistical methods ([Morris et al., 2019](#)) and may suffer from selective reporting ([Bouma et al., 2026](#); [Pawel et al., 2024, 2025](#)). One aspect of credibility that remains especially underexamined for simulations is reproducibility (i.e., the ability to reproduce the results of a study based on provided data, code, materials, and software/hardware) ([Hardwicke et al., 2020](#); [Montoya & Anderson, 2026](#)). In a simulation study this involves more than reanalyzing a fixed dataset, because the code also generates the data. Reproducing such a study therefore entails reproducing the data generation, the analysis, and the summary of results. Montoya and Anderson ([2026](#)) conceptually address each of these stages separately.

Reproducibility in this sense is not all-or-nothing but occupies a continuum ([Peng, 2011](#)), on which a study is positioned according to the extent of the materials that are publicly available. This ranges from publication alone at the low end to code and data that are linked and directly executable at the high end. Reaching that executable end is often treated as *full* reproducibility ([Peikert et al., 2021](#); [Peikert & Brandmaier, 2021](#)), and open science initiatives have pushed work toward it ([Kidwell et al., 2016](#); [Levenstein & Lyle, 2018](#); [Nosek et al., 2015](#)). Yet code and data are never fully self-sufficient ([Epskamp, 2019](#); [Wiebels & Moreau, 2021](#); [Ziemann et al., 2023](#)). Even error-free code depends on a hierarchy of software components, including the programming language version, the packages used in the analysis, and the system libraries those packages rely on. When these differ from the originals, code may fail, behave inconsistently across machines, or yield conflicting numerical results ([Baker et al., 2024](#); [Glatard et al., 2015](#); [Hodges et al., 2023](#); [Nosek et al., 2022](#)). These problems are particularly acute for simulation studies, which typically rely on a large amount of interconnected code, packages versions, and the underlying system itself to generate their data and produce the results of interest ([Luijken et al., 2024](#); [Siepe et al., 2024](#)).

To make this concrete, we use *computational environment* to refer to the complete software context required for an analysis to run successfully: the programming language version, package versions, system libraries, and operating system (Figure 1) (Rodrigues, 2023; Rodrigues & Baumann, 2026). We define *computational environment reproducibility* as the ability to reconstruct this entire set of software dependencies on any machine and at any future time, such that executing the same code yields the same numerical results<sup>1</sup>. Empirical assessments show that current practice falls short of this ideal. Indeed, Siepe et al. (2024) report that, among simulation studies published in psychology in 2021 and 2022, nearly two-thirds provide no accompanying code, and among those that do, documentation of the computational environment is rarely included. This gap is consequential, as simulation studies inform methodological recommendations, meaning that insufficient reproducibility undermines confidence in those recommendations (Luijken et al., 2024; White et al., 2024).

These challenges may persist for several reasons. Arguably, one is that even researchers aware of these demands face a fragmented landscape of solutions, each addressing only part of the problem. Package-level managers such as {renv} (Ushey, 2024) and {groundhog} (Simonsohn, 2020) stabilize R package versions but do not manage the R interpreter itself or the system-level libraries those packages depend on. Workflow orchestration tools such as {targets} (Landau, 2021) and Make (Feldman, 1979) support reproducibility in a different sense: they specify the structure of an analysis by formalizing the order in which steps should run and by tracking dependencies among intermediate results. These tools clarify *how* an analysis proceeds, but they assume that the software stack required to run each step is already stable. Containerization tools such as Docker, including R-focused implementations like Rocker (Boettiger, 2015; Boettiger & Eddelbuettel, 2017) offer a more comprehensive approach by bundling the full environment (i.e., operating system, system libraries, interpreter versions, and packages) into a single executable image. Yet their use requires familiarity with Linux system administration, and even

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<sup>1</sup> We note that the definition of computational environment reproducibility in this context may be considered rather loose. For the interested reader, we refer to Malka et al. (2026) (p. 3) and Lamb and Zacchiroli (2022).

containerization may suffer from temporal drift when Dockerfiles rely on mutable upstream repositories (Malka et al., 2026). For a detailed comparison of these tools and their limitations, see Rodrigues and Baumann (2026). Researchers thus face a difficult choice between solutions that are accessible but incomplete or approaches that are powerful but demand technical expertise.

In this article, therefore, we focus specifically on computational environment reproducibility as the foundation upon which other reproducibility practices depend. For that, we introduce Nix (Dolstra et al., 2004), a functional software<sup>2</sup> ecosystem designed to make software installation deterministic, and {rix} (Rodrigues & Baumann, 2025), an R interface that allows researchers to use Nix without needing deep knowledge of its underlying language or infrastructure. Our main objective with the tutorial is not to introduce a specific workflow orchestration system or to prescribe a particular analytic structure.<sup>3</sup> Instead, we aim to show what Nix and {rix} are and how they can establish a stable, cross-platform environment within which any simulation study—whether organized in simple, documented script sequences (e.g., .R files that `source()` others), through more formal orchestration tools (e.g., {targets}) or embedded as code chunks in .Rmd or .qmd—can be executed reliably.

We illustrate these ideas through a reproducible simulation study conducted in R, culminating in this automated APA-formatted manuscript generated with apaquarto (Schneider, 2024). Although the example centers on R because of its prominence in psychological methodology, the principles underlying environment reproducibility with {rix} apply equally to other languages, including Python and Julia, and to different development environments such as RStudio, VS Code, Emacs, or Positron. At the end of the article, we also briefly introduce {rixpress} (Rodrigues, 2025), a workflow orchestration tool that builds on Nix to coordinate

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<sup>2</sup> *Functional* is used in the sense of functional programming (pure functions, immutability, explicit inputs), not in the colloquial sense of *working*.

<sup>3</sup> We leave workflow orchestration aside for two reasons. First, many tools address it, each with its own conventions, and doing them justice would require a more tailored treatment than a single tutorial allows. Second, although formal orchestration may be necessary for full reproducibility, a study whose scripts are well documented, whose dependencies between files are explained, and whose environment is made stable arguably already sits high on the reproducibility spectrum. Moreover, a stable computational environment is what the other practices ultimately rest on, and it is what we focus on here. For readers interested in articles with a larger focus on the workflow, we suggest Peikert et al. (2021) and van Lissa et al. (2021).

pipelines across R, Python, and Julia.

### Setting the Stage: A Simulation Study and Its Reproducibility Risks

Imagine you have just been awarded a grant to conduct a large-scale simulation study. The study is designed to evaluate the performance of a statistical estimator under varying data-generating conditions (see Appendix A for full technical details). This tutorial is organized around this scenario. We use this example to ground our discussion into a typical methods section, but readers can follow the tutorial without engaging deeply with the simulation itself. In practice, simulation studies are typically organized into multiple component files. To mimic this practice, our simulation is organized into five sequential scripts: data generation (`01_data_generation.R`), model specification (`02_models.R`), simulation execution (`03_run_simulation.R`), performance metric calculation (`04_performance_metrics.R`), and results visualization (`05_plots.R`)<sup>4</sup>.

Now suppose a researcher attempts to reproduce the simulation run and results reported in the article. What might prevent them from obtaining identical outcomes, assuming error-free code? The natural first concern is the version of packages. Installing R packages at a later time may lead to errors if functions have been renamed or deprecated (e.g., `dplyr::cur_data()` changed to `pick()`), or, although usually discouraged, internal functions, such as `lavaan:::lav_utils_get_ancestors`, renamed to `lavaan:::lav_graph_get_ancestors`), or to subtly different results due to changes in default settings (e.g., `stringsAsFactors` defaulting to `FALSE` as of R 4.0). Beyond package versioning, many packages rely on system-level libraries that must be installed separately from R. Our simulation illustrates this dependency structure directly: the `{rvinecopulib}` package interfaces with a C++ backend and links against external libraries such as Boost, Eigen, and RcppThread (Nagler & Vatter, 2025).

The R language version introduces another layer of dependency. Code written for R 4.0

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<sup>4</sup> We note that, for the sake of simplicity, we included all code directly within this document as executable chunks in a single `.qmd` file. This entails that the simulation runs from start to finish, producing results and figures automatically. This approach would be impractical for many real-world simulation studies, which are often too computationally intensive. We return to this trade-off later in the tutorial as the solution is quite simple: one may simply run the simulation outside the `.qmd` file and load the results for the analyses in the same file.

may rely on syntax or functionality that is unavailable in earlier versions (e.g., the native pipe `|>` introduced in R 4.1). More subtly, changes to R's random number generation across major versions mean that identical code executed with the same seed can nevertheless produce different random sequences (Ottoboni & Stark, 2018). This is especially consequential for simulation studies, because the random draws are not incidental to the analysis but constitute the data itself. Indeed, a simulation study regenerates its data throughout the multiple runs. The random number generator is therefore part of the computational environment: reproducing the data requires not only the same seed but the same generator algorithm (e.g., Mersenne-Twister) and R version. Moreover, reproducing every estimate exactly can depend on lower-level components still, such as the numerical libraries R uses for matrix computations (e.g., BLAS/LAPACK). With respect to the latter, we do not mean that this exact level of identity must be met. Reproducing the broader pattern of results is arguably enough. However, as we will mention, the potential solution proposed here increases the chances of such bit-for-bit reproducibility, consistent with empirical findings (Malka et al., 2026).

Finally, when analyses are embedded in a literate programming workflow (i.e., documents that combine narrative text and executable code; dynamic document generation) additional layers of software dependencies arise. For example, rendering R Markdown (`.rmd`) or Quarto documents (`.qmd`) requires both a document conversion tool (e.g., Pandoc, which converts `.rmd` or `.qmd` files into formats such as PDF or HTML) and a typesetting system such as a LaTeX distribution or Typst. Each of these components introduces its own versioning constraints and platform-specific installation requirements. Taken together, these layers highlight that reproducibility depends not only on code and data, but also on the broader computational environment in which analyses are executed.

### **Nix and {rix}: A Comprehensive Solution**

A potential solution to this problem is Nix (Dolstra et al., 2004), a software ecosystem whose primary concern is reproducible, declarative builds. To achieve this Nix is a programming language, a build tool and a package manager. This article focuses mostly on Nix the package



manager. Most package managers (think of Apple’s or Android’s app stores) are imperative: they modify a system’s state as they install or update software. Nix, in contrast, treats build instructions and dependencies as immutable, enforcing reproducible, declarative, and isolated environments across platforms. This allows researchers to specify exactly which versions of programming languages and packages an analysis requires, while Nix automatically resolves and pins the underlying system libraries, so the environment can be recreated reliably on any machine.

### Core Principles

Rather than installing software into global directories (e.g., `/usr/lib`), Nix places every package in its own directory under `/nix/store`. Each package path contains a cryptographic hash representing its precise inputs: source code, dependencies, and build instructions. Because these paths are content-addressed, multiple versions of the same software can coexist without conflict. A researcher can, for example, maintain projects requiring R 4.1.0 and R 4.3.3 side by side, or use different package versions across analyses ([Rodrigues & Baumann, 2025](#)).

The Nix ecosystem is built around `nixpkgs`, a version-controlled repository comprising more than 120,000 packages, including nearly all of CRAN and Bioconductor. By pinning a specific commit or date, researchers freeze the entire software stack (i.e., R itself, R packages, and all system libraries). This eliminates the system-dependency problems that tools like `{renv}` cannot address ([Rodrigues & Baumann, 2025](#)). Empirical work has shown strong rebuildability and reproducibility rates for historical `nixpkgs` snapshots ([Rodrigues & Baumann, 2026](#)), and binary caches, which often allow environments to materialize in seconds, make Nix practical for interactive research workflows ([Rodrigues & Baumann, 2025](#)).

### The {rix} Package: R Interface to Nix

As previously stated, Nix is also a functional programming language. Because the Nix package manager is declarative, it requires expressions written in this language to install software. However, since this language is unfamiliar to most researchers, we recommend using `{rix}` to lower this barrier. The `{rix}` package provides an R-native interface: a single call to `rix()` generates complete Nix configurations from standard R syntax, specifying R versions, CRAN

packages, system libraries, and even Python or Julia components when required. Users never need to read or write Nix code directly, as {rix} translates automatically (Rodrigues & Baumann, 2025).

A key feature of {rix} is its integration with rstats-on-nix, a community-maintained fork offering daily CRAN snapshots and weekly tested environments on Linux and macOS. Researchers can request, for example, `rix(date = "2024-12-14")` to obtain a validated and reproducible environment without manually assessing compatibility. After the configuration is generated, `nix_build()` instantiates the environment, and binary caches typically allow this to complete within seconds (Rodrigues & Baumann, 2025).

Although Nix is capable of replacing tools like Docker for isolation or {renv} for package management, it does not require an all-or-nothing transition. Researchers can adopt it gradually and use it alongside familiar tooling. For instance, by building Docker images with Nix, converting existing {renv} lockfiles, or running {targets} pipelines within a Nix-defined environment (Rodrigues & Baumann, 2025). This allows Nix to strengthen reproducibility while preserving established workflows. We will come back to this after the tutorial.

### **A Practical Example: Setting up a Reproducible Simulation Study with {rix}**

We now proceed from the lens of a researcher that wants to make their simulation study and findings reproducible. We then show how another researcher would be able to reproduce such study.

#### **Step 1: Installing Nix and {rix}**

Before proceeding, both Nix and the {rix} R package need to be installed. The quick start below covers the common case, but the complete, up-to-date instructions for each operating system are maintained in the {rix} documentation:

- **Linux and Windows (WSL2):**

<https://docs.ropensci.org/rix/articles/setting-up-linux-windows.html>

- **macOS:** <https://docs.ropensci.org/rix/articles/setting-up-macos.html>

The installation is the same across operating systems, with one prerequisite on Windows:

Nix runs inside the Windows Subsystem for Linux 2 (WSL2), which must be installed and configured first (see the {rix} documentation linked above). On all systems, install Nix by running the Determinate Systems installer in a terminal<sup>5</sup>. To open one: on macOS, go to Applications → Utilities → Terminal. For Windows, launch your WSL2 (Ubuntu) shell from the Start menu. For Linux, open your distribution's terminal application. Then run:

```
curl --proto '=https' --tlsv1.2 -sSf \  
-L https://install.determinate.systems/nix | \  
sh -s -- install
```

Once Nix is installed<sup>6</sup>, there are two ways to access {rix}, depending on whether R is already installed on your system. In this tutorial, we proceed as if R was already installed<sup>7</sup> (Listing 1):

---

**Listing 1** Installing {rix} from CRAN or developmental version

---

```
# CRAN version  
install.packages("rix")  
# Development version  
install.packages(  
  "rix",  
  repos = c(  
    "https://ropensci.r-universe.dev"  
  )  
)
```

---

<sup>5</sup> A *terminal* is a text-based way to give your computer instructions. Readers new to the command line may find the following link a useful primer: <https://swcarpentry.github.io/shell-novice/>. This was also suggested in Wiebels and Moreau (2021)

<sup>6</sup> It is worth noting that {rix} can generate Nix expressions even without Nix installed on your system: you can write a default.nix file without Nix, but you cannot build or enter the resulting environment unless Nix is installed (Rodrigues & Baumann, 2025).

<sup>7</sup> We, however, recommend uninstalling your local R and letting Nix manage R, R packages, and other tools entirely. This approach avoids potential conflicts between system-installed and Nix-managed software, an issue we will illustrate later in this tutorial. See <https://docs.ropensci.org/rix/articles/setting-up-linux-windows.html#case-1-you-dont-have-r-installed-and-wish-to-install-it-using-nix-as-well> for more details.

## Step 2: Specifying the Computational Environment

We then create a folder for this project (here named `Why-risk-it-when-you-can-rix-it`), where we have the `article.qmd` inside the `Manuscript` subfolder and the `.R` files inside the `Simulation` subfolder. After that, we establish a reproducible environment by creating a script that will generate the environment specification. We recommend creating a file named `generate-env.R` (or similar) in the project directory. This script will use the `rix()` function from the `{rix}` package to produce a `default.nix` file: a declarative specification that precisely defines all software dependencies required for the project.

In our case, where we use literate programming for generating the manuscript, we implement the following environment specification, which can be found on the GitHub repository (<https://github.com/felipelfv/Why-risk-it-when-you-can-rix-it>) as a file named `gen-env.R` (Listing 2):

As usually happens, one does not know in advance all the packages needed. This is not a problem. Researchers may add the desired packages in the arguments above “on-the-go”, but must always rebuild the Nix environment, as we will illustrate below. Moreover, note that we have more than just the R packages specified for the simulation scripts. This happens because we also included what is needed for the manuscript generation, not solely for the simulation code. In Appendix B, we mention more specifically the reasons for adding each package in `r_pkgs()` and `tex_pkgs()`. For now, we focus more on clarifying the different arguments for the `rix()` function.

### *The Environment Generation Script*

The `rix()` function<sup>8</sup> constructs this specification through a series of parameters that collectively describe the computational environment. Each parameter serves a distinct purpose in defining the environment’s characteristics.

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<sup>8</sup> For an overarching information on the function `rix()`, we suggest the following `{rix}` documentation: <https://docs.ropensci.org/rix/articles/project-environments.html>

---

**Listing 2** Environment specification for the manuscript using `rix()`

---

```
library(rix)

# choose the path to your project
project_directory <- "." # here, the "Why-risk-it-when-you-can-rix-it" folder

rix(
  date = "2026-01-14", # recommendation: pin a date
  r_pkgs = c(
    "rix", "quarto", "knitr", "marginaleffects",
    "simhelpers", "ggplot2", "doParallel", "doRNG", "cowplot",
    "dplyr", "svglite", "rvinecopulib"
  ), # the R packages you need
  system_pkgs = c("quarto"), # non-R system tools
  tex_pkgs = c("amsmath", "ninecolors", "apa7", "scalere1",
    "threeparttable", "threeparttablex", "endfloat", "environ",
    "multirow", "tcolorbox", "pdfcol", "tikzfill", "fontawesome5",
    "framed", "newtx", "fontaxes", "xstring", "wrapfig", "tabularray",
    "siunitx", "fvextra", "geometry", "setspace", "fancyvrb",
    "anyfontsize"
  ), # LaTeX packages for the manuscript
  ide = "rstudio", # editor to include
  project_path = project_directory,
  overwrite = TRUE
)
```

---

**Specifying the R version.** Researchers must first determine which version of R to use. This can be accomplished in two ways: The `r_ver` argument accepts an exact version string (e.g., “4.3.3”) or special designations such as “latest-upstream” for the most recent stable release. Alternatively, the `date` argument specifies a particular date (e.g., “2024-11-15”), which ensures that R and all packages correspond to the versions available on that date. The date-based approach is generally preferable for reproducibility, as it captures a complete snapshot of the R ecosystem at a single point in time. For this tutorial, as shown on top, we use the `date` parameter to ensure temporal consistency across all software components (Rodrigues & Baumann, 2025) (see {rix} documentation for more: <https://docs.ropensci.org/rix/articles/project-environments.html>).

**Declaring R package dependencies.** The `r_pkgs` argument accepts a character vector listing all required R packages by their CRAN names. These packages will be installed from the version repository corresponding to the specified date or R version. It is important to list all packages that the analysis will load directly; dependencies of these packages are automatically resolved by Nix. For packages requiring specific versions not corresponding to the chosen date, researchers can specify exact versions using the syntax "packagename@version" (e.g., "ggplot2@2.2.1"). For packages available only on GitHub or other Git repositories, the `git_pkgs` argument accepts a list structure containing repository URLs and specific commit hashes. For example:

---

**Listing 3** Example for the `git_pkgs` argument

---

```
git_pkgs = list(  
  package_name = "marginaleffects",  
  repo_url = "https://github.com/vincentarelbundock/marginaleffects",  
  commit = "304bff91dc31ae28b227a8485bfa4f7bdc86d625"  
)
```

---

This ensures that exact development versions are obtained (Rodrigues & Baumann, 2025). For our simulation study, all packages were used with their CRAN versions (see {rix} documentation for more details: <https://docs.ropensci.org/rix/articles/installing-r-packages.html>).

**Including system-level dependencies.** Many R-based workflows require tools beyond R packages. The `system_pkgs` parameter specifies system-level software such as Quarto for document generation, Git for version control, or Pandoc for document conversion. Critically, we include Quarto as a system package because this tutorial demonstrates computational reproducibility not merely of the simulation code, but of the complete manuscript itself. Our manuscript uses the `apaquarto` extension for APA formatting, stored in the project's `_extensions/` directory (Rodrigues & Baumann, 2025) (see {rix} documentation for more: <https://docs.ropensci.org/rix/articles/installing-system-tools.html>).

**Specifying LaTeX packages.** The `tex_pkgs` parameter specifies LaTeX packages needed for PDF compilation. When any packages are listed, Nix automatically includes a minimal TeXLive distribution (`scheme-small`) as a base, to which the specified packages are added. Determining the required LaTeX packages may involve some trial and error: Quarto's error messages during, for example, the PDF rendering indicate which packages are missing, and these can then be added to `tex_pkgs` (see {rix} documentation for more:

<https://docs.ropensci.org/rix/articles/installing-system-tools.html>). The packages included in the above code offer a nice starting point for researchers wanting to create manuscripts using LaTeX.

**Configuring the development environment.** The `ide` parameter controls whether an integrated development environment (IDE) is included in the Nix environment, allowing users to interactively develop and run code within their editor of choice. When `ide` is specified, the project can be opened directly in the corresponding IDE, with all dependencies provided by the Nix environment. For example, setting `ide = "rstudio"` installs a project-specific version of RStudio inside the Nix environment. This is required for RStudio because, unlike most other editors, it cannot attach to an external Nix shell unless it is itself installed via Nix. On macOS, RStudio is only available through Nix for R versions 4.4.3 or later (or environments dated 2025-02-28 or later); for earlier R versions, alternative editors must be used. Other supported IDEs include Positron (`ide = "positron"`), Visual Studio Code (`ide = "code"`), and command-line interfaces such as Radian (`ide = "radian"`). These tools may either be installed directly within the Nix environment using the `ide` parameter, or users may rely on an existing system installation by setting `ide = "none"` (or `ide = "other"`) and configuring `direnv` to automatically load the Nix environment when the project directory is opened<sup>9</sup>. All IDEs installed via Nix are project-specific and do not interfere with system-wide installations. Detailed configuration instructions are provided in the {rix} documentation:

<https://docs.ropensci.org/rix/articles/configuring-ide.html>

---

<sup>9</sup> `direnv` is a lightweight utility that integrates with the user's shell and automatically loads project-specific environment settings when navigating into a directory (via a `.envrc` file), and unloads them when leaving. This makes environment activation implicit and reduces the risk of running analyses in the wrong software context.

**Setting file output parameters.** The `project_path` parameter indicates where the `default.nix` file should be written (“.” denotes the current directory), while `overwrite` controls whether an existing file should be replaced. Adding to this, setting `print = TRUE`, which is another argument, displays the generated specification in the console for immediate verification (Rodrigues & Baumann, 2025).

### Generating the Environment Specification

After defining the computational environment, the `rix()` function must be executed to generate the `default.nix` file. This can be done interactively by running `rix()` in an R console (Listing 2). The file is written to the folder given in `project_path`, so you can confirm it succeeded by checking that this folder now contains `default.nix`. The resulting `default.nix` file serves as the complete environment specification and contains all information required to recreate the project in a fully reproducible manner.

### Step 3: Building and Using the Reproducible Environment

Once Step 2 is complete, build the reproducible environment by navigating to the study directory in a terminal. You may use either the integrated Terminal in RStudio (Tools → Terminal → New Terminal) or an external system terminal from which you are running the Nix project, as highlighted before. From the study directory, run the following command (Listing 4):

---

#### Listing 4 Building the Nix environment

---

```
# in the terminal, from the project directory:  
nix-build
```

---

The expected output should look similar to (Listing 5):

This command builds the environment according to the specification. The first execution will download and install all required packages, which may take a few minutes depending on network speed and system resources. Subsequent builds use cached packages and complete in seconds. Upon successful completion, a path to the constructed environment in the Nix store is printed (here, `/nix/store/qa7fq20m2f94szsnqzciwv8h4n81w43v-nix-shell`), and a



---

**Listing 5** Expected output from `nix-build`

---

```
unpacking 'https://github.com/rstats-on-nix/nixpkgs/archive/2025-08-25.tar.gz'
  into the Git cache...
warning: ignoring untrusted substituter...
warning: ignoring the client-specified setting...
/nix/store/qa7fq20m2f94szsnqzciwv8h4n81w43v-nix-shell
```

---

symbolic link named `result` appears in the project directory pointing to this location.

Note that the warnings indicate that you are not configured as a trusted user, so Nix cannot use the `rstats-on-nix` binary cache and will instead compile packages from source, which is slower.

To enable binary caching, see the `{rix}` documentation:

<https://docs.ropensci.org/rix/articles/binary-cache.html>.

To activate the environment, run (Listing 6):

---

**Listing 6** Activating the Nix environment

---

```
# in the terminal, from the project directory:
nix-shell
```

---

The expected output (if you have configured yourself as a trusted user, otherwise the same warnings will appear) should look similar to (Listing 7):

---

**Listing 7** Expected output from `nix-shell`

---

```
unpacking 'https://flakehub.com/f/DeterminateSystems/nixpkgs-weekly/...'
  into the Git cache...
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$
```

---

This command drops the user into a shell where all specified packages and tools are available. The shell prompt changes to indicate that a Nix environment is active (here, `[nix-shell:~/Why-risk-it-when-you-can-rix-it]$`). To verify that R is being provided by Nix rather than a system installation, run `which R`. This should return a path within `/nix/store/`. Moreover, from within the Nix shell, users can launch their IDE by typing its name

(e.g., `rstudio` or `positron`), which opens the IDE with the Nix environment active (Listing 8)<sup>10</sup>

---

**Listing 8** Activating RStudio

---

```
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$ rstudio
```

---

**Reproducing the Complete Manuscript**<sup>11</sup>

Within the nix shell, one is able to render the manuscript as follows in the terminal (Listing 9):

---

**Listing 9** Rendering the manuscript with Quarto

---

```
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$  
quarto render Manuscript/article.qmd
```

---

This command executes all code chunks in the manuscript, incorporates results (e.g., Table 1) and figures (e.g., Figure 1), and generates a formatted PDF following APA style guidelines via the `apaquarto` extension (Schneider, 2024). This extension is saved in the project repo already. To add it to a project of your own, run the following inside the Nix shell, where Quarto is provided by the environment and therefore needs no separate installation (Listing 10):

---

**Listing 10** Adding the `apaquarto` extension

---

```
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$  
quarto add wjschne/apaquarto
```

---

The final document (`.docx`, `.pdf`, or `.html`) is saved directly in the project folder<sup>12</sup>. Because Quarto is installed as a system-level package in our Nix specification, the rendering occurs

---

<sup>10</sup> Please note that activating an RStudio instance via Nix does **not** automatically open the specific project directory you are working in. We therefore recommend creating an **RStudio project file** (`.Rproj`) and opening that file when using Nix to ensure that RStudio is correctly associated with the intended project and environment.

<sup>11</sup> See the {rix} documentation for more: <https://docs.ropensci.org/rix/articles/literate-programming.html>

<sup>12</sup> Although we use `apaquarto` in this example, many alternative manuscript templates are available, and Nix is agnostic to the specific template employed, provided the necessary extensions are installed.

entirely within a fully reproducible environment, ensuring consistent output across machines regardless of local software configurations. If desired, the manuscript can also be reproduced interactively by opening the project folder in the user’s preferred IDE.

At this point, it is also worth noting that Nix shells do not fully isolate you from your existing system by default (as mentioned in footnote 7). For R users, this has a practical implication: packages installed in your regular R library (outside of Nix) could potentially be loaded when running R from within the Nix environment. The `{rix}` package addresses this automatically—when you call `rix()`, it also executes `rix_init()`, which creates a project-specific `.Rprofile`. This file configures R to ignore external package libraries and also disables `install.packages()` within the environment. The rationale is straightforward: any new packages should be added to `default.nix` and the environment rebuilt, preserving full reproducibility (Rodrigues & Baumann, 2025). However, for stricter isolation<sup>13</sup> that also prevents access to other system programs not specified in `default.nix`, use the `--pure` flag (Listing 11):

---

**Listing 11** Activating the Nix environment with strict isolation

---

```
nix-shell --pure
```

---

### *Reproducing the Simulation and Results*

As previously mentioned, researchers may prefer not to use literate programming, or embedding the simulation within a dynamic document may be impractical. In this case, one could still follow the same steps shown thus far focusing only on the `.R` files while still benefiting from a reproducible computational environment. For example, `03_run_simulation.R` begins by loading the required packages and sourcing other scripts that are needed:

As an illustration, if we were to focus solely on the computational reproducibility of the code underlying the simulation and results, the environment specification would be considerably

---

<sup>13</sup> For example, when preparing this manuscript without the `--pure` flag, `quarto render` worked successfully. However, when using the `--pure` flag, the build failed. Running `quarto check` from within the Nix shell (i.e., `nix-shell --run "quarto check"`) revealed that Quarto was still accessing the system’s LaTeX installation (`/Library/TeX/texbin`) rather than being restricted to only what was specified in `default.nix`.

---

**Listing 12** Code for running simulation

---

```
library(marginaleffects)
...

# Source helper functions
source("Simulation/01_data_generation.R")
...
```

---

simpler (Listing 13):

---

**Listing 13** Environment specification for simulation-only computational reproducibility

---

```
rix(
  date = "2026-01-14",
  r_pkgs = c(
    "simhelpers", "ggplot2", "doParallel", "doRNG", "cowplot", "dplyr",
    "rvinecopulib", "marginaleffects"
  ),
  ide = "rstudio",
  project_path = ".",
  overwrite = TRUE
)
```

---

After following Steps 1-3 with this simpler specification, the simulation study may be reproduced within the Nix shell (Listing 14) as follows:

---

**Listing 14** Running the complete simulation workflow

---

```
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$
Rscript Simulation/03_run_simulation.R
```

---

In the same way, we could proceed with 04\_performance\_metrics.R (Listing 15), which loads the simulation results (in sim\_results.rds) and calculates the performance metrics:

Similarly, 05\_plots.R (Listing 16) uses those saved metrics (in performance\_summary.rds) to create the plots:

Therefore, the key advantage of executing within nix-shell is that all dependencies (i.e.,

---

**Listing 15** Running the performance metrics calculation

---

```
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$  
Rscript Simulation/04_performance_metrics.R
```

---

---

**Listing 16** Generating the visualization plots

---

```
[nix-shell:~/Why-risk-it-when-you-can-rix-it]$  
Rscript Simulation/05_plots.R
```

---

R version, packages, and system tools) match exactly those specified in `default.nix`. Note, however, that this approach relies on manually running scripts in sequence. It ensures a reproducible environment but does not formalize the workflow itself; dependencies between scripts remain implicit in the code rather than explicitly declared.

**Additional Considerations for Advanced Workflows**

Thus far, we have presented Nix and {rix} as standalone solutions for computational reproducibility, contrasting initially them with tools like {renv}, Docker, and {targets}. However, these tools are not mutually exclusive, as in many cases, they can complement each other (Rodrigues & Baumann, 2025). Additionally, as highlighted in previous literature (Peikert & Brandmaier, 2021; Piccolo & Frampton, 2016; Siepe et al., 2024), fully reproducible research benefits not only from a stable computational environment but also from explicit workflow orchestration. We therefore briefly introduce {rixpress}, which extends Nix-based reproducibility to formalized, multi-language pipelines. These topics are not covered in depth; our goal is simply to clarify how these tools relate to one another and orient readers toward resources for more advanced use cases.

***Workflow Orchestration: {targets} and {rixpress}***

Complex simulation studies often benefit from workflow management systems that track dependencies between computational steps, cache intermediate results, and enable selective re-execution when inputs change. Two complementary approaches exist within the Nix ecosystem.

**Using {targets} Within Nix.** The {targets} package (Landau, 2021) provides workflow orchestration for R-based projects. To integrate {targets} with Nix, include targets in the `r_pkgs` parameter of `rix()` and execute the pipeline within `nix-shell` using `Rscript -e 'targets::tar_make()'` . A shell hook can also be added via the `shell_hook` argument to run the pipeline automatically when entering the Nix shell. This approach is ideal for projects that remain within R and do not require different environments for different pipeline steps (see {rix} documentation: <https://docs.ropensci.org/rix/articles/reproducible-pipelines.html>).

**Using {rixpress} for Multi-Language Pipelines.** The {rixpress} package (Rodrigues, 2025), a sister package to {rix}, uses Nix itself as the build automation tool rather than operating within a Nix environment. Each pipeline step becomes a Nix derivation, built in isolation with automatic caching based on content. The key advantage emerges in multi-language workflows: different steps can execute in different Nix-defined environments (e.g., one step using a specific version of R, another using Python, another using Julia). The interface, inspired by {targets}, uses functions like `rxp_r()`, `rxp_py()`, and `rxp_jl()` to define pipeline steps (see {rixpress} documentation: <https://docs.ropensci.org/rixpress/articles/intro-concepts.html>). The GitHub repository for this article directs interested readers to a demonstration of {rixpress} applied to this entire project.

### ***Multi-Language Environment Support***

While this tutorial focuses on R, researchers working across multiple programming languages can include Python or Julia in their environments. The `py_conf` parameter accepts a list specifying a Python version and required packages (Listing 17). Similarly, `jl_conf` enables Julia package installation. This capability is particularly useful, for example, for projects requiring statistical computing in R alongside machine learning pipelines in Python or numerical optimization in Julia (Rodrigues & Baumann, 2025) (see {rix} documentation for more: <https://docs.ropensci.org/rix/articles/installing-r-packages.html>).

---

**Listing 17** Including Python packages

---

```
py_conf = list(py_version = "3.12", py_pkgs = c("polars", "pandas"))
```

---

***Converting Existing {renv} Projects***

Researchers with existing {renv} projects can migrate using the `renv2nix()` function, which reads an `renv.lock` file and generates an equivalent Nix expression. This is particularly valuable for projects where {renv} encountered system dependency issues or where stricter reproducibility guarantees are desired. Unlike {renv}, which captures R package versions but not the R interpreter or system libraries, Nix manages all layers of the software stack (see {rix} documentation: <https://docs.ropensci.org/rix/articles/renv2nix.html>).

***Containerization with Docker***

Nix and Docker are not necessarily mutually exclusive (Rodrigues & Baumann, 2026). Researchers already using Docker do not need to abandon it to benefit from Nix—the two can be combined by using Nix inside Docker containers to handle environment setup (Rodrigues & Baumann, 2025). This is particularly useful for deployment to cloud platforms or high-performance computing clusters where Docker is standard but Nix may not be available (see {rix} documentation: <https://docs.ropensci.org/rix/articles/nix-inside-docker.html>).

**Discussion**

Reproducibility in computational research is often treated as a matter of transparency—making data and code available. This tutorial has argued that transparency alone is insufficient without the ability to reliably reconstruct the computational environments in which analyses are executed. For simulation studies in particular, where results depend critically on software versions, system libraries, and random number generation, environment-level reproducibility is not optional but essential.

By introducing Nix and the {rix} package, we demonstrated a practical and accessible approach to fully specifying and rebuilding computational environments for simulation-based

research. This approach enables analyses and manuscripts to be rerun identically across machines and over time, transforming reproducibility from an aspirational goal into a verifiable property of the research workflow.

Importantly, adopting environment reproducibility does not require abandoning existing analytic practices. Nix is agnostic to programming language, editor, workflow structure, and manuscript template, allowing researchers to retain familiar tools while strengthening the reliability of their work. In this sense, reproducible environments serve as enabling infrastructure—supporting, rather than replacing, other best practices such as version control, workflow orchestration, and transparent reporting.

If reproducibility is to function as a cornerstone of cumulative science, then the ability to reconstruct computational environments must become a routine part of methodological practice. Tools such as Nix and {rix} lower the barrier to achieving this goal, making fully reproducible simulation research feasible without requiring deep systems expertise. We hope this tutorial helps normalize environment-level reproducibility as a standard component of rigorous computational research in psychology and beyond.



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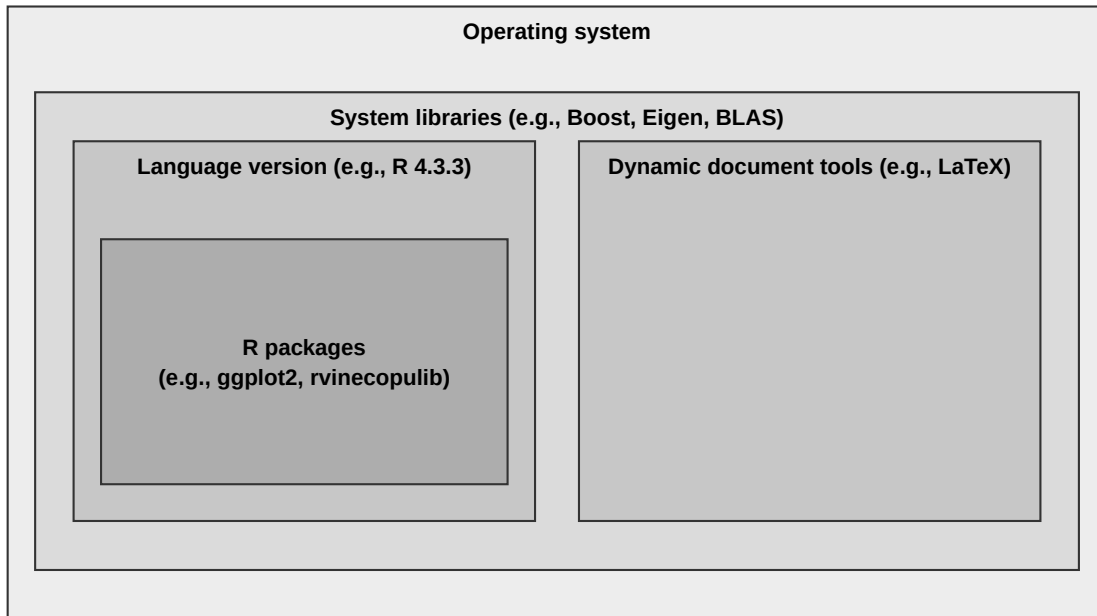
**Table 1**

*Performance metrics for ACE estimator across simulation conditions. Values in parentheses are Monte Carlo standard errors (MCSE).*

| Sample Size | Confounding | Relative Bias | Relative RMSE | Coverage      | CI Width      |
|-------------|-------------|---------------|---------------|---------------|---------------|
| 50          | none        | 0.957 (0.042) | 0.422 (0.035) | 0.910 (0.029) | 0.224 (0.004) |
| 100         | none        | 1.013 (0.026) | 0.256 (0.020) | 0.930 (0.026) | 0.159 (0.002) |
| 2000        | none        | 0.997 (0.006) | 0.059 (0.005) | 0.960 (0.020) | 0.036 (0.000) |
| 50          | mild        | 1.173 (0.036) | 0.395 (0.034) | 0.890 (0.031) | 0.222 (0.004) |
| 100         | mild        | 1.157 (0.028) | 0.318 (0.026) | 0.870 (0.034) | 0.154 (0.002) |
| 2000        | mild        | 1.114 (0.006) | 0.130 (0.006) | 0.510 (0.050) | 0.035 (0.000) |
| 50          | severe      | 1.200 (0.044) | 0.482 (0.054) | 0.870 (0.034) | 0.223 (0.005) |
| 100         | severe      | 1.235 (0.030) | 0.379 (0.027) | 0.830 (0.038) | 0.157 (0.002) |
| 2000        | severe      | 1.257 (0.007) | 0.266 (0.007) | 0.020 (0.014) | 0.035 (0.000) |

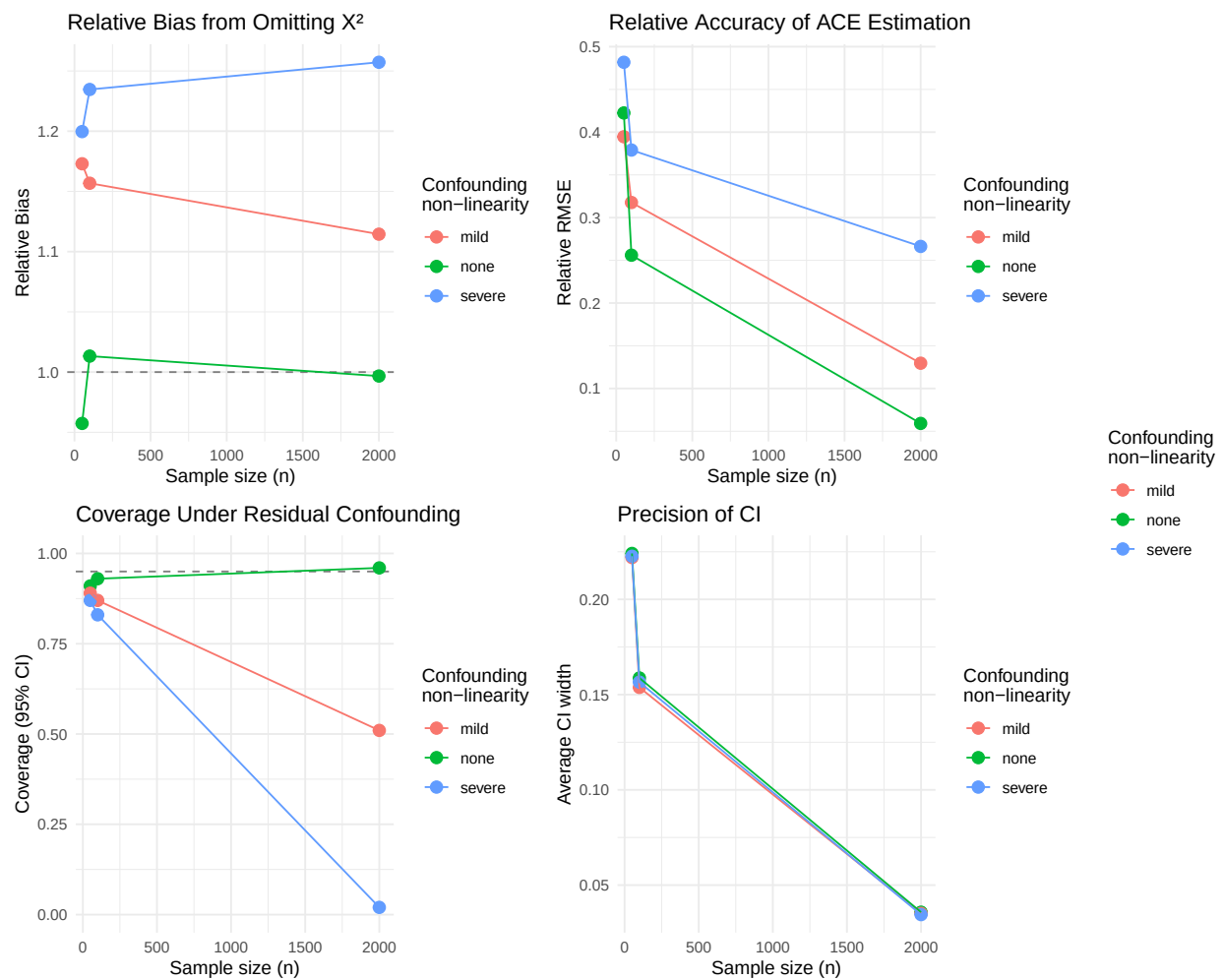
**Figure 1**

*The computational environment as a set of nested layers, based on Peikert & Brandmaier (2021).*



**Figure 2**

Performance of ACE estimator across sample sizes and confounding severity. Panel A shows relative bias, Panel B shows relative RMSE, Panel C shows coverage probability of 95% confidence intervals (dashed line at nominal 0.95 level), and Panel D shows average confidence interval width. Results demonstrate that model misspecification induces systematic bias that persists across sample sizes, while increasing sample size improves precision but not accuracy under misspecification.





## Appendix A

### Simulation Study Design

Here we present a rather short description following recommendations from previous research, but ideally even more may be reported (Morris et al., 2019; Pawel et al., 2025; Siepe et al., 2024; White et al., 2024). This mimics a methods or similar section in articles.

**Factorial Design.** The simulation employs a full factorial design with two factors: sample size ( $n \in \{50, 100, 2000\}$ ) and degree of confounding non-linearity ( $\gamma_2 \in \{0, 0.3, 0.8\}$ , labeled as none, mild, and severe). The parameter  $\gamma_2$  controls the strength of the quadratic confounder effect on the outcome (see Data Generation). This yields nine conditions, each replicated  $K = 100$  times.

**Data Generation.** For each replication, data are generated following a causal structure where a confounder  $X_2$  affects both treatment assignment and the outcome. The confounder and treatment error term are generated using the {rvinecopulib} package: pairs  $(U_1, U_2)$  are drawn from an independence copula via `rbicop()`, then transformed to standard normals via  $X_2 = \Phi^{-1}(U_1)$  and  $\epsilon = \Phi^{-1}(U_2)$ . The independence copula is simply  $C(u, v) = uv$ , meaning the resulting uniforms are independent—mathematically equivalent to calling `rnorm()` directly. We use {rvinecopulib} intentionally because it depends on C++ libraries.

Treatment assignment follows  $X_1 = \alpha_0 + \alpha_1 X_2 + \alpha_2 X_2^2 + \epsilon$  where  $\alpha_0 = 0$ ,  $\alpha_1 = 0.5$ , and  $\alpha_2 = 0.2$ . This creates confounding because  $X_2$  influences treatment assignment through both linear and quadratic terms. The binary outcome is generated from the true logistic regression model:

$$\text{logit}(P(Y = 1 \mid X_1, X_2)) = \beta_0 + \beta_1 X_1 + \gamma_1 X_2 + \gamma_2 X_2^2$$

with parameters  $\beta_0 = -0.5$ ,  $\beta_1 = 0.7$  (the causal effect of interest),  $\gamma_1 = -0.4$ , and  $\gamma_2$  varying by condition. The analyst misspecifies the outcome model by omitting the quadratic confounder term, fitting instead:

$$\text{logit}(P(Y = 1 \mid X_1, X_2)) = \beta_0 + \beta_1 X_1 + \gamma_1 X_2$$

This misspecification creates residual confounding because the omitted term  $\gamma_2 X_2^2$  is

correlated with  $X_1$  (since  $X_1$  depends on both  $X_2$  and  $X_2^2$ ), violating the conditional exchangeability assumption given linear adjustment alone.

**Estimand.** The target estimand is the average causal effect (ACE) of  $X_1$  on  $Y$ , properly adjusted for confounding:

$$\text{ACE}(X_1) = \mathbb{E} \left[ \frac{\partial P(Y = 1 \mid X_1, X_2)}{\partial X_1} \right] = \mathbb{E} \left[ \beta_1 \cdot \frac{\exp(\eta)}{(1 + \exp(\eta))^2} \right]$$

where  $\eta = \beta_0 + \beta_1 X_1 + \gamma_1 X_2 + \gamma_2 X_2^2$  is the correctly specified linear predictor, and the expectation is taken over the joint distribution of  $(X_1, X_2)$ . For each  $\gamma_2$  condition, the “true” ACE (denoted  $\theta$ ) is approximated once using a very large sample ( $N = 200,000$ ) with the correctly specified model including  $X_2^2$ .

**Estimator.** The causal effect is estimated from the misspecified model (omitting  $X_2^2$ ) as:

$$\widehat{\text{ACE}}(X_1) = \frac{1}{n} \sum_{i=1}^n \tilde{\beta}_1 \cdot \frac{\exp(\tilde{\eta}_i)}{(1 + \exp(\tilde{\eta}_i))^2}$$

where  $\tilde{\eta}_i = \tilde{\beta}_0 + \tilde{\beta}_1 X_{i1} + \tilde{\gamma}_1 X_{i2}$  and  $\tilde{\beta} = (\tilde{\beta}_0, \tilde{\beta}_1, \tilde{\gamma}_1)$  are maximum likelihood estimates from the misspecified logistic regression.

**Performance Criteria.** Table B1 presents the performance criteria used to evaluate the ACE estimator across simulation conditions.

**Computational Details.** The simulation was conducted on a MacBook Pro (Apple M4 Pro Chip), running macOS Sequoia 15.7.3. All analyses were performed in R (version 4.5.2). Parallel processing was implemented through the {doParallel} package [version 1.0.17; Corporation and Weston (2022)], with {doRNG} [version 1.8.6.2; Gaujoux (2025)] to ensure independent and reproducible random number streams. For data generation we used the {rvinecopulib} package [version 0.7.3.1.0; Nagler and Vatter (2025)]. The estimator was implemented using the {marginaleffects} package [version 0.31.0; Arel-Bundock et al. (2024)]. Data wrangling was performed with {dplyr} [version 1.1.4; Wickham et al. (2023)]. Method performance was assessed through multiple metrics following the formulas from the {simhelpers}

package [version 0.3.1; Joshi and Pustejovsky (2025)]. Figures were produced with {ggplot2} [version 4.0.1; Wickham (2016)] and {cowplot} [version 1.2.0; Wilke (2025)].

## Appendix B

### Clarifying the packages used

#### R Packages

##### *Reproducibility Infrastructure*

**rix**: Generates the Nix expression (`default.nix`) for reproducible environments.

##### *Simulation Study*

These packages are used in the `Simulation_Scripts/` folder:

**rvinecopulib** (used in `01_data_generation.R`): Generates correlated data via copulas using the `rbicop()` function.

**marginaleffects** (used in `02_models.R`): Computes average causal effects via the `avg_slopes()` function.

**doParallel** (used in `03_run_simulation.R`): Enables parallel foreach loops across CPU cores.

**doRNG** (used in `03_run_simulation.R`): Makes parallel RNG reproducible.

**simhelpers** (used in `04_performance_metrics.R`): Calculates bias, RMSE, and coverage metrics.

**ggplot2** (used in `05_plots.R`): Creates simulation result visualizations.

**cowplot** (used in `05_plots.R`): Combines plots with `plot_grid()` and extracts legends.

**dplyr** (used in `article.qmd`): Data wrangling for results reported in Table 1.

##### *Dynamic Document Generation*

**quarto**: R interface to invoke Quarto rendering.

**knitr**: Executes R code chunks in `.qmd` files.

**svglite**: SVG graphics device; `apaquarto` sets `dev: svglite` for HTML output.

#### LaTeX Packages

##### *Required by apaquarto Extension*

These are loaded in `apaquarto's header.tex` or `apatemplate.tex`:

**amsmath**: Math environments (align, equation, etc.).

**threeparttablex**: Tables with notes below (APA table format).

**tcolorbox**: Callout boxes (note, warning, tip blocks).

**fontawesome5**: Icons in callouts.

**multirow**: Table cells spanning multiple rows.

**newtx**: Times-like fonts (default when no custom mainfont).

**geometry**: Page margins.

### *Dependencies of apaquarto Packages*

**environ**: Dependency of tcolorbox.

**pdfcol**: Dependency of tcolorbox.

**tikzfill**: Dependency of tcolorbox.

**fontaxes**: Dependency of newtx.

**xstring**: Template conditionals.

**scalereel**: Dependency of apa7 class.

### *Required by apa7 Document Class*

**apa7**: The document class itself (\documentclass{apa7}).

**endfloat**: Moves floats to end of document in manuscript mode.

**threeparttable**: Dependency for threeparttablex.

### *Required by Quarto PDF Rendering*

**framed**: Shaded/framed regions for callouts.

**fvextra**: Enhanced verbatim for syntax-highlighted code blocks.

**fancyvrb**: Verbatim environments for code display.

**setspace**: Line spacing; also used in article.qmd for single-spaced code blocks.

**anyfontsize**: Arbitrary font sizes in code blocks.

### *Additional Packages*

**ninecolors**: Extended color palettes.

**wrapfig:** Text wrapping around figures.

**tabularray:** Modern table typesetting.

**siunitx:** SI units and number formatting.

**Table B1**

*Performance criteria for evaluating the ACE estimator.  $\hat{\theta}_k$  denotes the ACE estimate from replication  $k$  (for  $k = 1, \dots, K$ ), where  $K = 1000$  is the number of replications, and  $\theta$  denotes the true ACE for a given condition. For coverage and width criteria,  $A_k$  and  $B_k$  denote the lower and upper endpoints of the 95% confidence interval from replication  $k$ ,  $W_k = B_k - A_k$  is the interval width,  $c_\beta$  is the estimated coverage probability, and  $I(\cdot)$  is an indicator function equaling 1 if the condition is true and 0 otherwise. The Monte Carlo standard error (MCSE) quantifies the simulation uncertainty in each performance measure estimate*

| Criterion     | Estimate                                                                       | MCSE                                                                                                                       |
|---------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Bias          | $\frac{1}{K} \sum_{k=1}^K \hat{\theta}_k - \theta$                             | $\sqrt{\frac{1}{K(K-1)} \sum_{k=1}^K (\hat{\theta}_k - \bar{\hat{\theta}})^2}$                                             |
| Variance      | $\frac{1}{K-1} \sum_{k=1}^K (\hat{\theta}_k - \bar{\hat{\theta}})^2$           | $\sqrt{\frac{K-1}{K} \cdot \frac{1}{K-1} \sum_{k=1}^K (\hat{\theta}_k - \bar{\hat{\theta}})^2}$                            |
| RMSE          | $\sqrt{\frac{1}{K} \sum_{k=1}^K (\hat{\theta}_k - \theta)^2}$                  | $\sqrt{\frac{K-1}{K} \sum_{j=1}^K \left( \sqrt{(\hat{\theta}_j - \theta)^2} - \overline{RMSE} \right)^2}$                  |
| Relative Bias | $\frac{1}{\theta K} \sum_{k=1}^K \hat{\theta}_k$                               | $\frac{1}{\theta} \sqrt{\frac{1}{K(K-1)} \sum_{k=1}^K (\hat{\theta}_k - \bar{\hat{\theta}})^2}$                            |
| Relative RMSE | $\frac{1}{\theta} \sqrt{\frac{1}{K} \sum_{k=1}^K (\hat{\theta}_k - \theta)^2}$ | $\frac{1}{\theta} \sqrt{\frac{K-1}{K} \sum_{j=1}^K \left( \sqrt{(\hat{\theta}_j - \theta)^2} - \overline{RMSE} \right)^2}$ |
| Coverage      | $\frac{1}{K} \sum_{k=1}^K I(A_k \leq \theta \leq B_k)$                         | $\sqrt{\frac{c_\beta(1-c_\beta)}{K}}$                                                                                      |
| Width         | $\frac{1}{K} \sum_{k=1}^K (B_k - A_k)$                                         | $\sqrt{\frac{1}{K(K-1)} \sum_{k=1}^K (W_k - \bar{W})^2}$                                                                   |